# SamaSetu

Live Hindi → Tamil speech translation, in the speaker's own voice

Great Indian AI Internship Challenge 2026

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**I don't know a word of Tamil, yet SamaSetu lets me speak it in my own voice, about 1.5 seconds after I finish a Hindi sentence, with no English in between.**




Live screen: Hindi heard by Gnani Prisma (left), Tamil spoken in the cloned voice by Gnani Timbre (right), delay measured per sentence. Sentences from a real test run.

## THE PROBLEM

Millions of Hindi speakers live, work and get treated in Tamil Nadu. At a pharmacy or hospital counter, language becomes a wall.

Most tools route through English. Hindi and Tamil are both **SOV**, English is **SVO**, so the sentence is reordered twice, and the voice sounds like a machine.

## THE SOLUTION

- **Direct Hindi → Tamil**: no English pivot, one word order throughout.
- **Your own voice**: cloned from a 20-second Hindi sample, speaking Tamil.
- **Live**: ~1.5 s to first sound, on a laptop CPU.
- **Two-way mode** for real conversations.

## HOW IT WORKS

01 · HEAR
**Gnani Prisma v2.5**
Realtime streaming speech-to-text over WebSocket
**~110 ms**

→

02 · TRANSLATE
**IndicTrans2 320M**
Direct Hindi → Tamil, local CPU, greedy decoding
**~395 ms**

→

03 · SPEAK
**Gnani Timbre v2.5**
Cloned-voice TTS, streamed over SSE, plays from the first chunk
**~407 ms**

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**~1.5 s**
speech to first translated sound

## MEASURED, NOT ASSUMED

| STAGE | SLOWER OPTION | CHOSEN |
|---|---|---|
| Speech-to-text | ~~680 ms (upload)~~ | **110 ms (stream)** |
| Translation | ~~1,211 ms (5 beams)~~ | **395 ms (greedy)** |
| Voice output | ~~2,712 ms (full file)~~ | **407 ms (stream)** |

## WHO IT'S FOR

Patients & health workers · Migrant workers · Shopkeepers · Support agents

## GNANI AI MODELS USED

Prisma v2.5 · streaming STT
Timbre v2.5 · voice clone
Timbre v2.5 · cloned-voice TTS

## HONEST LIMITS

Waits for a pause (per sentence). Clear sentences work best; slang can mistranslate. Not yet tested on phone audio.

## WHAT'S NEXT

Speak while the person is still talking; more language pairs; **IndicSimul**, the missing benchmark for live Indic-to-Indic speech translation.

**SamaSetu**: "sama" (equal) + "setu" (bridge). Language should be a bridge, not a wall.

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